

1 Problem 1

Part a)

Let S_{10} denote the total number of girls among the 10 newborns. Since each birth independently results in a girl with probability $p = \frac{1}{2}$, then S_n follows a binomial distribution (Definition 1.2.9c):

$$S_{10} \sim \text{Bin}(10, \tfrac{1}{2}),$$

with probability mass function

$$\mathbb{P}(S_{10} = k) = \binom{10}{k} \left(\frac{1}{2}\right)^{10}, \quad k = 0, 1, \dots, 10.$$

At least 60% of 10 newborns being girls means $S_{10} \geq 0.6 \times 10 = 6$. Hence

$$a_{10} = \mathbb{P}(S_{10} \geq 6).$$

Using the complement and the additivity of $\mathbb{P}$ (Lemma 1.1.10b), we compute directly:

$$\begin{aligned} a_{10} &= \sum_{k=6}^{10} \mathbb{P}(S_{10} = k) = \sum_{k=6}^{10} \binom{10}{k} \left(\frac{1}{2}\right)^{10} = \frac{1}{2^{10}} \sum_{k=6}^{10} \binom{10}{k} \\ &= \frac{1}{1024} \left(\binom{10}{6} + \binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right) \\ &= \frac{1}{2^{10}} (210 + 120 + 45 + 10 + 1) = \frac{386}{1024} = \frac{193}{512} \\ &\approx 0.3770. \end{aligned}$$

Part b)

We claim $a_{100} < a_{10}$ and prove it.

Proof. Let $X_1, \dots, X_{100}$ be independent, identically distributed random variables, where $X_k = 1$ if the k -th newborn is a girl and $X_k = 0$ otherwise. Then

$$S_{100} := \sum_{k=1}^{100} X_k \sim \text{Bin}(100, \tfrac{1}{2}),$$

and the event that at least 60% of 100 newborns are girls means $S_{100} \geq 60$. Hence

$$a_{100} = \mathbb{P}(S_{100} \geq 60).$$

We have that (by Example 3.3.14a and 3.3.36):

$$\begin{aligned} \mathbb{E}(S_{100}) &= np = 100 \cdot \frac{1}{2} = 50, \\ \text{Var}(S_{100}) &= np(1-p) = 100 \cdot \frac{1}{2} \cdot \frac{1}{2} = 25. \end{aligned}$$

We observe that the event $\{S_{100} \geq 60\}$ implies $S_{100} - 50 \geq 10$, which in turn implies $|S_{100} - 50| \geq 10$. Therefore

$$\{S_{100} \geq 60\} \subseteq \{|S_{100} - 50| \geq 10\}.$$

By monotonicity of $\mathbb{P}$ (Lemma 1.1.10d), the subset relation $\{S_{100} \geq 60\} \subseteq \{|S_{100} - 50| \geq 10\}$ implies:

$$a_{100} = \mathbb{P}(S_{100} \geq 60) \leq \mathbb{P}(|S_{100} - 50| \geq 10).$$

Since $\mathbb{E}(S_{100}) = 50$ and $\text{Var}(S_{100}) = 25 < \infty$ (which ensures $S_{100} \in L^2(\Omega, \mathcal{F}, \mathbb{P})$), we can apply Chebyshev's inequality (Theorem 3.3.25) with $\varepsilon = 10$ to bound this probability:

$$\begin{aligned}\mathbb{P}(|S_{100} - 50| \geq 10) &= \mathbb{P}(|S_{100} - \mathbb{E}(S_{100})| \geq 10) \\ &\leq \frac{\text{Var}(S_{100})}{10^2} = \frac{25}{100} = \frac{1}{4} \\ \implies a_{100} &\leq \frac{1}{4}.\end{aligned}$$

From part a) we know that $a_{10} = \frac{193}{512}$. Comparing:

$$a_{10} = \frac{193}{512} \approx 0.377 > 0.25 = \frac{1}{4} \geq a_{100}.$$

Hence $a_{100} < a_{10}$. □

Part c)

Let $X_1, X_2, \dots$ be independent, identically distributed Bernoulli($\frac{1}{2}$)-random variables as before, and let

$$M_n := \frac{S_n}{n} = \frac{1}{n} \sum_{k=1}^n X_k.$$

The condition “at least 60% of n newborns are girls” means $S_n \geq \frac{3}{5}n$, or equivalently $M_n = \frac{S_n}{n} \geq \frac{3}{5}$, so

$$a_n = \mathbb{P}\left(M_n \geq \frac{3}{5}\right).$$

We have $\mathbb{E}|X_k| = \frac{1}{2} \leq 1 < \infty$ (Example 3.3.14), so $X_k \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ (Definition 3.3.8 (ii)). Therefore, we can apply the 2nd Weak Law of Large Numbers (Theorem 4.3.4), which gives:

$$M_n \xrightarrow{p} \mathbb{E}(X_1) = \frac{1}{2}.$$

We observe that the event $\{M_n \geq \frac{3}{5}\}$ implies $M_n - \frac{1}{2} \geq \frac{3}{5} - \frac{1}{2} = \frac{1}{10}$, which in turn implies $|M_n - \frac{1}{2}| \geq \frac{1}{10}$. Therefore

$$\left\{M_n \geq \frac{3}{5}\right\} \subseteq \left\{\left|M_n - \frac{1}{2}\right| \geq \frac{1}{10}\right\}.$$

By monotonicity of $\mathbb{P}$ (Lemma 1.1.10d), we can bound this using a strict inequality for any $0 < \varepsilon < \frac{1}{10}$ (for example, $\varepsilon = 0.09$). Applying convergence in probability (Definition 4.1.1) yields:

$$0 \leq a_n = \mathbb{P}\left(M_n \geq \frac{3}{5}\right) \leq \mathbb{P}\left(\left|M_n - \frac{1}{2}\right| > \varepsilon\right) \xrightarrow{n \rightarrow \infty} 0.$$

Hence $\lim_{n \rightarrow \infty} a_n = 0$. □